Toward Reliability in the NISQ Era: Robust Interval Guarantee for Quantum Measurements on Approximate States

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Introduction

We present a method to guarantee the reliability of readout from NISQ devices. The promises of many quantum algorithms rely crucially on the ability to perform accurate measurements and obtain expectation values of observables. However, particularly in the NISQ era, these devices are prone to noise, decoherence and imperfect implementations which lead to unreliable measurement outcomes. In the absence of an abundance of qubits for implementing error correction codes, this ultimately undermines the reliability of these algorithms and hence fundamentally questions their utility in the near term. It is thus crucial to have methods which allow us to accurately quantify these errors. In this work, we take a first step towards this goal and present intervals which are guaranteed to contain the true expectation value of an observable under an ideal, but unavailable target quantum state σ, using only an approximation ρ of this state.

Ideal evolution **(unavailable)**

$$|0\rangle \xrightarrow{U} |\psi\rangle \xrightarrow{\boxed{A}} \langle H\rangle_\psi$$

Noisy circuit **(available)**

$$|0\rangle \xrightarrow{\tilde{U}} \rho \xrightarrow{\boxed{A}} \langle H\rangle_\rho$$

$$\Longrightarrow \quad |\langle H\rangle_\rho - \langle H\rangle_\psi| \leq \delta$$

Robustness Interval via Quantum Hypothesis Testing

Mathematical Tool: Quantum Hypothesis Testing

Goal: discriminate between the null hypothesis ρ and the alternative σ via a POVM $\{\Lambda, \mathbb{1} - \Lambda\}$ while minimizing the error probabilities

$$\text{Type-I:} \quad \alpha(\Lambda;\, \rho) := \mathrm{Tr}\,[\Lambda\rho]\,, \qquad \text{Type-II:} \quad \beta(\Lambda;\, \sigma) := \mathrm{Tr}\,[(\mathbb{1} - \Lambda)\sigma]$$

One way to do this, is to minimize the type-II error β under the constraint that the type-I error α is below some threshold α_0

$$\beta^*(\alpha_0;\, \rho,\, \sigma) := \inf_{0 \leq \Lambda \leq \mathbb{1}} \{\beta(\Lambda;\, \sigma) :\, \alpha(\Lambda;\, \rho) \leq \alpha_0\} \tag{1}$$

Lemma 1. *Let $\sigma,\, \rho \in \mathcal{S}(\mathcal{H}_d)$ be arbitrary quantum states, $\alpha_0 \in [0,\, 1]$ and suppose that $\mathcal{F}(\rho,\, \sigma) \geq 1 - \epsilon$ for $\epsilon \leq 1 - \alpha_0$. We have*

$$\beta^*(\alpha_0;\, \rho,\, \sigma) \;\geq\; \alpha_0 + (1-\epsilon)(1 - 2\alpha_0) - 2\sqrt{\epsilon(1-\epsilon)\alpha_0(1-\alpha_0)} \tag{2}$$

with equality if the states σ and ρ are pure and $\mathcal{F}(\rho,\, \sigma) = 1 - \epsilon$.

Robustness Interval

For any $0 \leq A \leq \mathbb{1}$, we have

$$\begin{aligned} \mathrm{Tr}\,[A\sigma] \;&\leq\; \sup_{\Lambda}\{\mathrm{Tr}\,[\Lambda\sigma] \;:\; 0 \leq \Lambda \leq \mathbb{1},\, \mathrm{Tr}\,[\Lambda\rho] = \mathrm{Tr}\,[A\rho]\} \\ &= 1 - \beta^*(\mathrm{Tr}\,[A\rho]\,;\, \rho,\, \sigma). \end{aligned} \tag{3}$$

Combining this with Lemma 1 leads to the following result:

Theorem 1. *Let $\sigma,\, \rho$ be quantum states with $\mathcal{F}(\rho,\, \sigma) \geq 1 - \epsilon$ for some $\epsilon \geq 0$. Let A be an observable with $-\mathbb{1} \leq A \leq \mathbb{1}$. Then, for $\epsilon \leq \frac{1}{2}(1 + \langle A\rangle_\rho)$, we have the lower bound*

$$\langle A\rangle_\sigma \;\geq\; (1 - 2\epsilon)\langle A\rangle_\rho - 2\sqrt{\epsilon(1-\epsilon)(1 - \langle A\rangle_\rho^2)} \tag{4}$$

Similarly, for $\epsilon \leq \frac{1}{2}(1 - \langle A\rangle_\rho)$, we have the upper bound

$$\langle A\rangle_\sigma \;\leq\; (1 - 2\epsilon)\langle A\rangle_\rho + 2\sqrt{\epsilon(1-\epsilon)(1 - \langle A\rangle_\rho^2)} \tag{5}$$

Robustness Intervals based on Gram Matrices

Mathematical Tool: Gram Determinants

Pioneered by Weinhold [1], the Gramian method uses the fact that for a Hermitian operator A, and states $|\psi\rangle, |\phi\rangle$, the Gram matrix $G(|\psi\rangle,\, |\phi\rangle,\, A|\phi\rangle)$ is non-negative

$$G(|\psi\rangle,\, |\phi\rangle,\, A|\phi\rangle) = \begin{pmatrix} 1 & \langle\psi|\phi\rangle & \langle\psi|A|\phi\rangle \\ \langle\phi|\psi\rangle & 1 & \langle\phi|A|\phi\rangle \\ \langle\phi|A|\psi\rangle & \langle\phi|A|\phi\rangle & \langle\phi|A^2|\phi\rangle \end{pmatrix} \geq 0 \tag{6}$$

⇒ Restricts values that $\langle\psi|A|\phi\rangle$ can take to be within the roots of the determinant of G, $\langle\psi|A|\phi\rangle \in \langle\phi|\psi\rangle\langle A\rangle_\phi \pm \Delta A_\phi\sqrt{1 - |\langle\phi|\psi\rangle|^2}$

⇒ Extension to mixed states via purifications $|\Psi\rangle,\, |\Phi\rangle$ with $\mathcal{F}(\rho,\, \sigma) = |\langle\Psi|\Phi\rangle|^2$

Lower Bound to Expectation Values

For any observable $A \geq 0$ and states $\rho,\, \sigma$ with $\mathcal{F}(\rho,\, \sigma) \geq 1 - \epsilon$, for $\epsilon \geq 0$ with $\sqrt{(1-\epsilon)/\epsilon} \geq \Delta_\rho A / \langle A\rangle_\rho$, we have

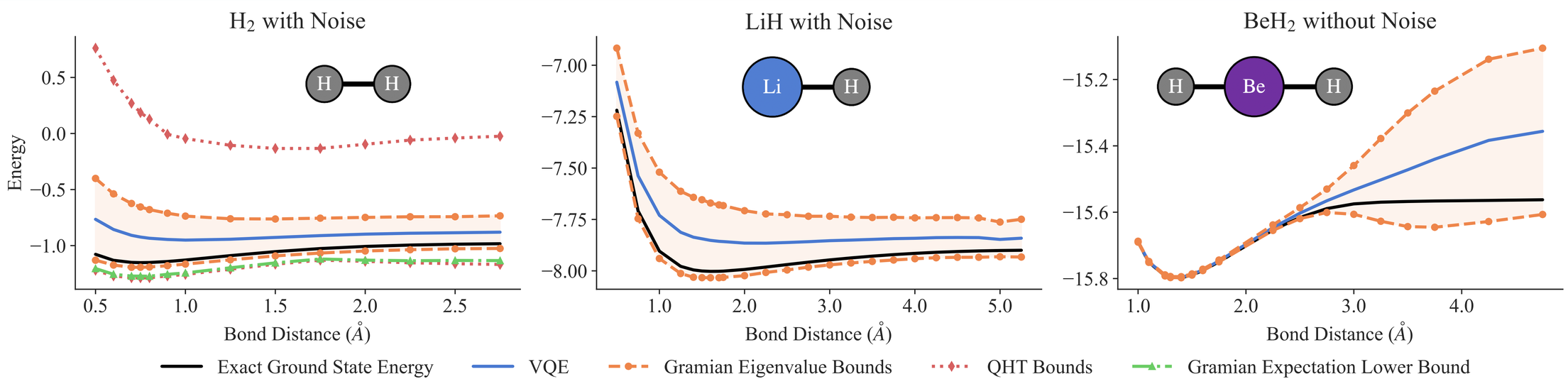
$$\langle A\rangle_\sigma \;\geq\; \langle A\rangle_\rho - 2\sqrt{\epsilon(1-\epsilon)}\Delta_\rho A - \epsilon\left(\langle A\rangle_\rho - \frac{(\Delta_\rho A)^2}{\langle A\rangle_\rho}\right) \tag{7}$$

Robustness Interval for Eigenvalues

For any observable A with eigenvalue $A|\psi\rangle = \lambda|\psi\rangle$, and a state ρ with fidelity $\langle\psi|\rho|\psi\rangle \geq 1 - \epsilon$, we have

$$\langle A\rangle_\rho - \Delta A_\rho\sqrt{\frac{\epsilon}{1-\epsilon}} \;\leq\; \lambda \;\leq\; \langle A\rangle_\rho + \Delta A_\rho\sqrt{\frac{\epsilon}{1-\epsilon}} \tag{8}$$

Numerical Simulations



References

[1] Frank Weinhold. Lower bounds to expectation values. *Journal of Physics A: General Physics*, 1(3):305, 1968.